NYPD Shooting Data Project

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Importing the Data

Let's begin by importing the data.

```
NYPD_Shooting_data <- read_csv("https://data.cityofnewyork.us/api/v3/views/833y-fsy8/export.csv")
```

Summary of Raw Data

I would like to look at a summary of the raw data. Unfortunately, the OCCUR_TIME field as a *<time>* class causes an issue, so we will exclude that from our summary:

```
NYPD_Shooting_data %>%
  select(-OCCUR_TIME) %>%
  summary()
```

```
##   INCIDENT_KEY          OCCUR_DATE          BORO       LOC_OF_OCCUR_DESC
##  Min.   :  9953245   Length   :29744   Length   :29744   Length   :29744
##  1st Qu.: 67321140   N.unique : 6431   N.unique :    5   N.unique :    2
##  Median :109291972   N.blank  :    0   N.blank  :    0   N.blank  :    0
##  Mean   :133850951   Min.nchar:   10   Min.nchar:    5   Min.nchar:    6
##  3rd Qu.:214741917   Max.nchar:   10   Max.nchar:   13   Max.nchar:    7
##  Max.   :299462478                                       NAs      :25596
##
##     PRECINCT      JURISDICTION_CODE LOC_CLASSFCTN_DESC   LOCATION_DESC
##  Min.   :  1.00   Min.   :0.0000    Length   :29744    Length   :29744
##  1st Qu.: 44.00   1st Qu.:0.0000    N.unique :   10    N.unique :   40
##  Median : 67.00   Median :0.0000    N.blank  :    0    N.blank  :    0
##  Mean   : 65.23   Mean   :0.3181    Min.nchar:    5    Min.nchar:    3
##  3rd Qu.: 81.00   3rd Qu.:0.0000    Max.nchar:   11    Max.nchar:   25
##  Max.   :123.00   Max.   :2.0000    NAs      :25596    NAs      :14977
##                   NAs    :2
##  STATISTICAL_MURDER_FLAG   PERP_AGE_GROUP       PERP_SEX         PERP_RACE
##  Mode :logical           Length   :29744   Length   :29744   Length   :29744
##  FALSE:23979             N.unique :   12   N.unique :    4   N.unique :    8
##  TRUE :5765              N.blank  :    0   N.blank  :    0   N.blank  :    0
##                          Min.nchar:    3   Min.nchar:    1   Min.nchar:    5
##                          Max.nchar:    7   Max.nchar:    6   Max.nchar:   30
##                          NAs      : 9344   NAs      : 9310   NAs      : 9310
##
##    VIC_AGE_GROUP        VIC_SEX           VIC_RACE       X_COORD_CD
```

```
##  Length   :29744   Length   :29744   Length   :29744   Min.   : 914928
##  N.unique :    7   N.unique :    3   N.unique :    7   1st Qu.:1000094
##  N.blank  :    0   N.blank  :    0   N.blank  :    0   Median :1007826
##  Min.nchar:    3   Min.nchar:    1   Min.nchar:    5   Mean   :1009442
##  Max.nchar:    7   Max.nchar:    1   Max.nchar:   30   3rd Qu.:1016739
##                                                       Max.   :1066815
##
##    Y_COORD_CD        Latitude       Longitude           Lon_Lat
##  Min.   :125757   Min.   :40.51   Min.   :-74.25   Length   :29744
##  1st Qu.:183042   1st Qu.:40.67   1st Qu.:-73.94   N.unique :14057
##  Median :195506   Median :40.70   Median :-73.91   N.blank  :    0
##  Mean   :208722   Mean   :40.74   Mean   :-73.91   Min.nchar:   24
##  3rd Qu.:239980   3rd Qu.:40.83   3rd Qu.:-73.88   Max.nchar:   45
##  Max.   :271128   Max.   :40.91   Max.   :-73.70   NAs      :   97
##                   NAs    :97      NAs    :97
```

Tidy and Transform the Data

Now we shall set the proper data types, remove columns we don't expect to be relevant for our analysis, and get things in a format we're comfortable analyzing.

Remove Irrelevant Columns

I am going to really trim down the data for ease of use for this project. Some of the information is very interesting, but for the purposes of this assignment, I do not expect to use it.

As an example, the **LOC_OF_OCCUR_DESC** could be an interesting stat to research. Looking at what shootings took place inside verses outside could be fascinating. However, with over 85% of the incidents being blank and not providing that information, I do not want to infer context based on less than 15% of the reported shootings.

I will not elaborate on all of the fields I am removing, but feel free to review that list in the code below.

```
NYPD_Shooting_data <- NYPD_Shooting_data %>%
  select(-c(LOC_OF_OCCUR_DESC,
            JURISDICTION_CODE,
            LOC_CLASSFCTN_DESC,
            LOCATION_DESC,
            JURISDICTION_CODE,
            LOC_CLASSFCTN_DESC,
            LOCATION_DESC,
            STATISTICAL_MURDER_FLAG,
            X_COORD_CD,
            Y_COORD_CD,
            Latitude,
            Longitude,
            Lon_Lat))
```

Clean Up Data

There are some fields that have erroneous data. The **PERP_AGE_GROUP** and **VIC_AGE_GROUP** field should ideally only have categories of *<18*, *18-24*, *25-44*, *45-65*, *65+*, and some variant of unknown or

NA. When looking at the data, there are also blanks/nulls along with noisy data entries like "224" or "940" that do not make sense as a person's age. There are similar inconsistencies in the sex fields.

For the purpose of our analysis, we will create a global standard of *NA* for the blanks and unknowns in those fields and massage the data that may have been initially entered incorrectly. Review the code below to see those changes.

```
allowed_ages <- c("<18", "18-24", "25-44", "45-65", "65+", "NA")
allowed_sexes <- c("M", "F", "NA")
NYPD_Shooting_data <- NYPD_Shooting_data %>%
  mutate(across(where(is.character), ~if_else(.x == "(null)", "NA", .x))) %>%
  mutate(across(where(is.character), ~if_else(.x == "UNKNOWN", "NA", .x))) %>%
  mutate(across(c(PERP_AGE_GROUP, VIC_AGE_GROUP),
                ~if_else(.x %in% allowed_ages, as.character(.x), "NA"))) %>%
  mutate(across(c(PERP_SEX, VIC_SEX),
                ~if_else(.x %in% allowed_sexes, as.character(.x), "NA")))
```

Rename Columns and Change Data Types

I am going to rename the remaining fields to my comfort along with making the data types more friendly for filtering, plotting, and analyzing. Review the following code for those changes.

```
NYPD_Shooting_data <- NYPD_Shooting_data %>%
  rename(ID = INCIDENT_KEY,
         Date = OCCUR_DATE,
         Time = OCCUR_TIME,
         Borough = BORO,
         Precinct = PRECINCT,
         PerpAge = PERP_AGE_GROUP,
         PerpSex = PERP_SEX,
         PerpRace = PERP_RACE,
         VictimAge = VIC_AGE_GROUP,
         VictimSex = VIC_SEX,
         VictimRace = VIC_RACE) %>%
  mutate(
    ID = as.character(ID),
    Date = mdy(Date),
    Borough = as.factor(Borough),
    Precinct = as.character(Precinct),
    PerpAge = as.factor(PerpAge),
    PerpSex = as.factor(PerpSex),
    PerpRace = as.factor(PerpRace),
    VictimAge = as.factor(VictimAge),
    VictimSex = as.factor(VictimSex),
    VictimRace = as.factor(VictimRace))
```

Summary of Cleaned Data

Now that we have gone through and cleaned the data, let's review a summary as it stands to compare to the raw data. The time field is still an issue, so we will exclude that from this summary as well:

```
NYPD_Shooting_data %>%
  select(-Time) %>%
  summary()
```

```
##          ID             Date                        Borough           Precinct
##  Length   :29744   Min.   :2006-01-01   BRONX         : 8834   Length   :29744
##  N.unique :23298   1st Qu.:2009-10-29   BROOKLYN      :11685   N.unique :   77
##  N.blank  :    0   Median :2014-03-25   MANHATTAN     : 3977   N.blank  :    0
##  Min.nchar:    7   Mean   :2014-10-31   QUEENS        : 4426   Min.nchar:    1
##  Max.nchar:    9   3rd Qu.:2020-06-29   STATEN ISLAND:  822   Max.nchar:    3
##                    Max.   :2024-12-31
##
##   PerpAge      PerpSex              PerpRace     VictimAge     VictimSex
##  <18  : 1805   F :  461   BLACK         :12323   <18  : 3081   F : 2891
##  18-24: 6630   M :16845   NA            : 3466   18-24:10677   M :26841
##  25-44: 6342   NA:12438   WHITE HISPANIC: 2667   25-44:13563   NA:   12
##  65+  :   67              BLACK HISPANIC: 1487   65+  :  236
##  NA   :14900              WHITE         :  305   NA   : 2187
##                           (Other)       :  186
##                           NAs           : 9310
##                           VictimRace
##  AMERICAN INDIAN/ALASKAN NATIVE:   13
##  ASIAN / PACIFIC ISLANDER      :  478
##  BLACK                         :20999
##  BLACK HISPANIC                : 2930
##  NA                            :   72
##  WHITE                         :  741
##  WHITE HISPANIC                : 4511
```

Comparing Raw and Cleaned Data Summaries The summary of the clean data already tells us so much more information than what we were able to decipher from the raw data summary. It is also important to make sure none of the data was lost or inadvertently manipulated during the tidying process. If this was a more serious, real-world presentation, I might review the counts of particular fields in more detail, but for the purposes of this assignment, I will not get that granular.

What I can tell from these summaries is that if you add totals of the individual fields like **Borough**, **Precinct**, etc, they all add up to 29,744, which is equal to the number of starting rows. Based on the minor changes we made to some field types and correcting noisy data, it gives me enough confidence that we have not lost or broken anything.

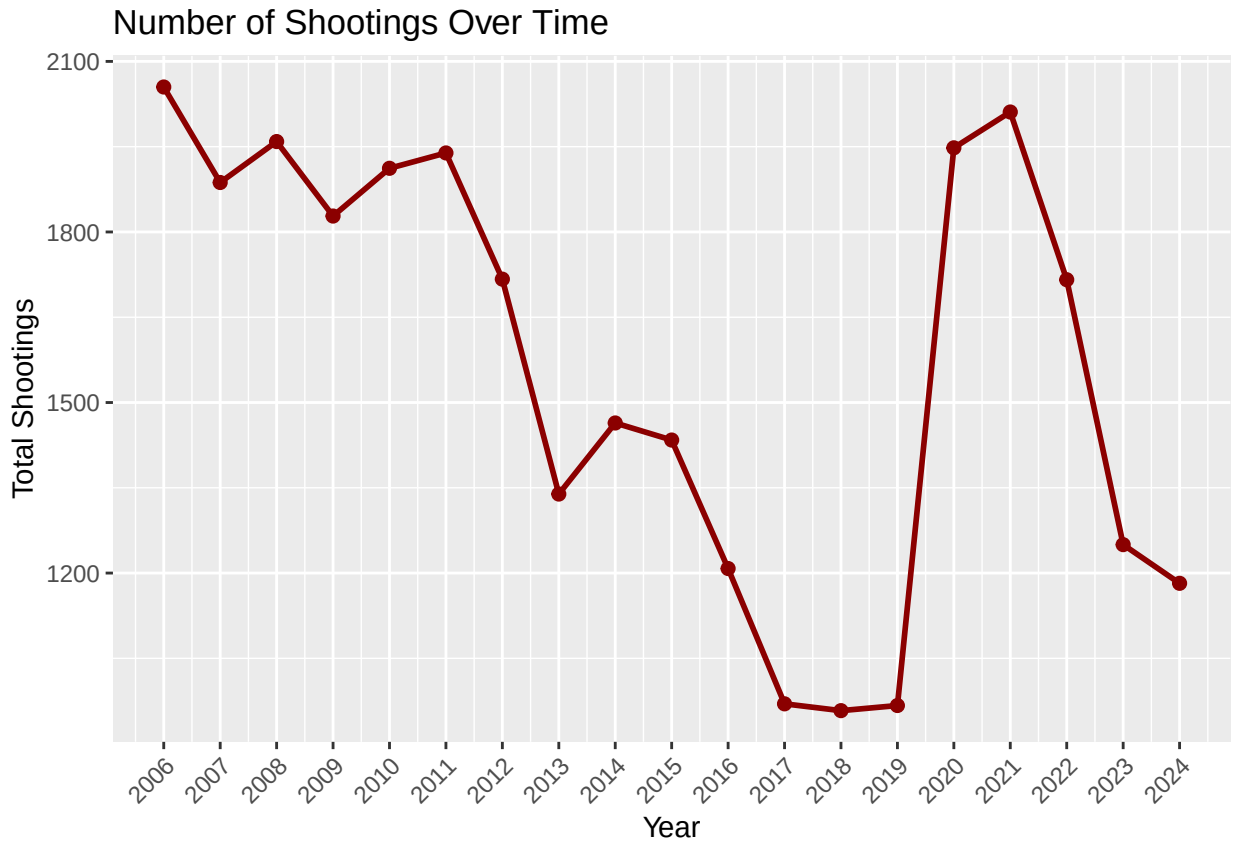
Visualizing the Data

With the clean data, let's examine some trends and see what we see.

Number of Shootings Over Time

This set of data covers 19 years of shootings in New York City from 2006 through 2024. I am initially curious to see the number of shootings over that course of time.

```r
yearly_counts <- NYPD_Shooting_data %>%
  mutate(Year = year(Date)) %>%
  count(Year)
ggplot(yearly_counts, aes(x = Year, y = n)) +
  geom_line(color = "darkred", linewidth = 1) +
  geom_point(color = "darkred", size = 2) +
  scale_x_continuous(breaks = seq(2006, 2024, by = 1))+
  labs(
    title = "Number of Shootings Over Time",
    x = "Year",
    y = "Total Shootings"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



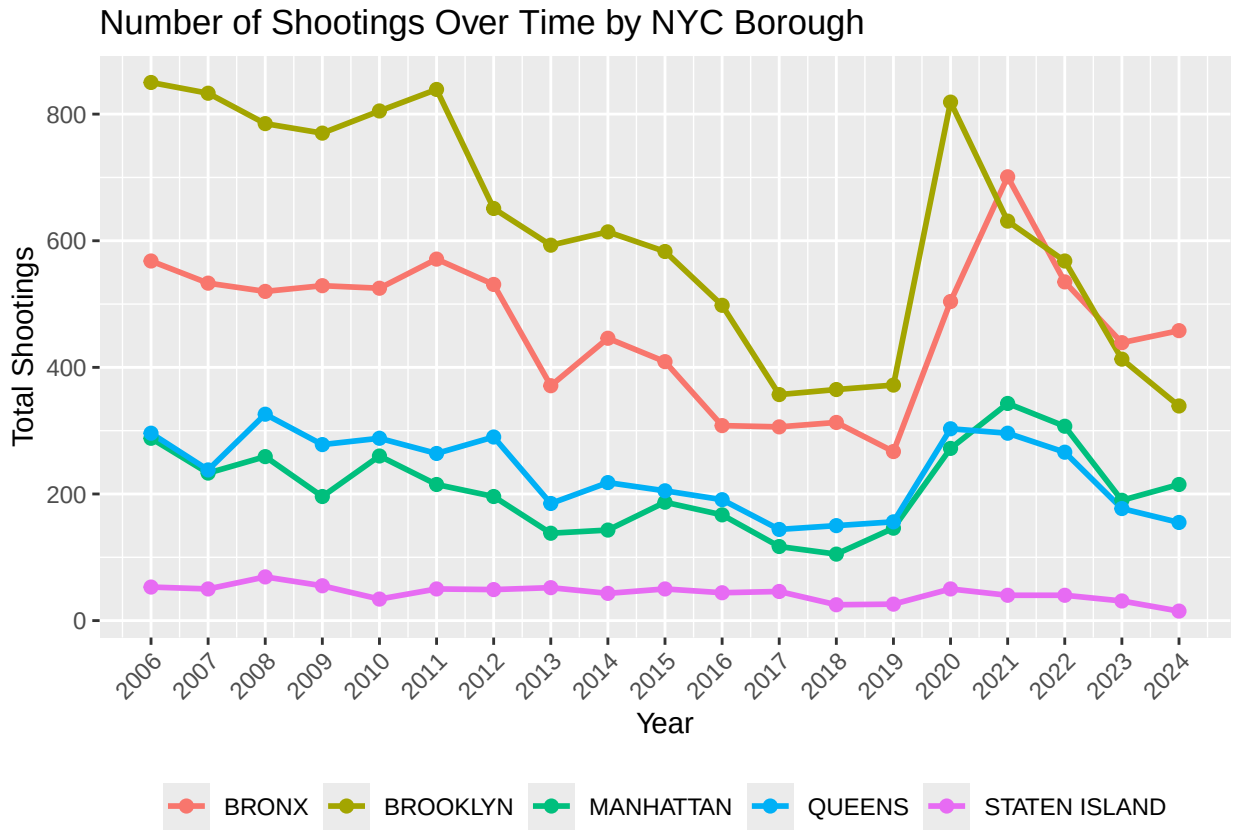
Analysis of Graph

The resulting graph shows an interesting trend. There was a big drop in gun violence in New York City from 2006 through 2019 before a massive rise at the start of the Covid-19 pandemic. There was yet another drop after the pandemic from 2021 up until recently. The data collected stops at the end of 2024, but I did search and find a recent article posted by NYC.gov supports the data shown by the graph. It also goes on to state shootings continued to drop to get close to or beat the low 2017-2019 records.

Adding in the Boroughs

This graph made me wonder if the shooting trend in New York City was mirrored among the different boroughs. Let me add those lines to this graph to see what the case is.

```
borough_yearly_counts <- NYPD_Shooting_data %>%
  mutate(Year = year(Date)) %>%
  count(Year, Borough)
ggplot(borough_yearly_counts, aes(x = Year, y = n, color = Borough, group = Borough)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = seq(2006, 2024, by = 1))+
  labs(
    title = "Number of Shootings Over Time by NYC Borough",
    x = "Year",
    y = "Total Shootings",
    color = NULL
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "bottom")
```



Analysis of Graph

The individual boroughs mostly follow the overall totals. Staten Island has been consistent with the number of shootings year over year. The other four boroughs have pairs that show similar behavior. It is interesting because the Bronx and Brooklyn are close in numbers, but they do not border each other geographically. I

would have expected those two to be next to each other. The similarities in numbers must be tied more to social, economic, or other characteristics not reflected in this data.

Modeling the Safety of Boroughs

A modeling opportunity I can imagine with the data points available is to map out the safest boroughs by a victim's age and time. The concept makes me think about my parents warning me not to go into the city late at night ... so let's see what the data tells us!

Constructing the Model

Our **Time** data was cleaned, but is still not great to work with. I will be breaking the different times down into factors where the day is split into 6-hour increments.

```
safety_matrix <- NYPD_Shooting_data %>%
  filter(!is.na(Borough), !is.na(VictimAge), !is.na(Time)) %>%
  mutate(Hour = as.numeric(Time)/3600, TimeBin = case_when(
    Hour >= 0 & Hour < 6 ~ "Night",
    Hour >= 6 & Hour < 12 ~ "Morning",
    Hour >= 12 & Hour < 18 ~ "Afternoon",
    Hour >= 18 & Hour < 24 ~ "Evening"),
    TimeBin = as.factor(TimeBin)) %>%
  count(Borough, VictimAge, TimeBin, name = "Shootings")
mod <- glm(Shootings ~ Borough * VictimAge * TimeBin, family = poisson(), data = safety_matrix)
summary(mod)
```

```
##
## Call:
## glm(formula = Shootings ~ Borough * VictimAge * TimeBin, family = poisson(),
##     data = safety_matrix)
##
## Coefficients:
##                                                  Estimate Std. Error z value
## (Intercept)                                      5.288267   0.071067  74.413
## BoroughBROOKLYN                                  0.525864   0.089641   5.866
## BoroughMANHATTAN                                -0.997808   0.136927  -7.287
## BoroughQUEENS                                   -0.755668   0.125711  -6.011
## BoroughSTATEN ISLAND                            -2.069391   0.212251  -9.750
## VictimAge18-24                                   0.978934   0.083355  11.744
## VictimAge25-44                                   1.196368   0.081100  14.752
## VictimAge65+                                    -1.822531   0.190527  -9.566
## VictimAgeNA                                     -0.318454   0.109521  -2.908
## TimeBinEvening                                   0.910212   0.084161  10.815
## TimeBinMorning                                  -1.677349   0.179102  -9.365
## TimeBinNight                                     0.381614   0.092189   4.139
## BoroughBROOKLYN:VictimAge18-24                  -0.068030   0.105516  -0.645
## BoroughMANHATTAN:VictimAge18-24                 -0.022369   0.160965  -0.139
## BoroughQUEENS:VictimAge18-24                     0.094269   0.146209   0.645
## BoroughSTATEN ISLAND:VictimAge18-24             -0.172458   0.254569  -0.677
## BoroughBROOKLYN:VictimAge25-44                  -0.158256   0.103050  -1.536
## BoroughMANHATTAN:VictimAge25-44                  0.085326   0.155171   0.550
```

```
## BoroughQUEENS:VictimAge25-44                          0.102915   0.142346   0.723
## BoroughSTATEN ISLAND:VictimAge25-44                  -0.544043   0.259635  -2.095
## BoroughBROOKLYN:VictimAge65+                         -0.354013   0.256128  -1.382
## BoroughMANHATTAN:VictimAge65+                        -0.070033   0.375377  -0.187
## BoroughQUEENS:VictimAge65+                           -0.630627   0.414793  -1.520
## BoroughSTATEN ISLAND:VictimAge65+                    -1.396345   1.037444  -1.346
## BoroughBROOKLYN:VictimAgeNA                          -0.269930   0.142676  -1.892
## BoroughMANHATTAN:VictimAgeNA                          0.446835   0.194272   2.300
## BoroughQUEENS:VictimAgeNA                            -0.039758   0.195275  -0.204
## BoroughSTATEN ISLAND:VictimAgeNA                     -0.127833   0.338371  -0.378
## BoroughBROOKLYN:TimeBinEvening                       -0.419893   0.109070  -3.850
## BoroughMANHATTAN:TimeBinEvening                       0.160621   0.159614   1.006
## BoroughQUEENS:TimeBinEvening                         -0.412373   0.156114  -2.641
## BoroughSTATEN ISLAND:TimeBinEvening                  -0.491501   0.270922  -1.814
## BoroughBROOKLYN:TimeBinMorning                       -0.308140   0.238331  -1.293
## BoroughMANHATTAN:TimeBinMorning                       0.220103   0.323419   0.681
## BoroughQUEENS:TimeBinMorning                         -0.552665   0.377929  -1.462
## BoroughSTATEN ISLAND:TimeBinMorning                  -0.848380   0.756358  -1.122
## BoroughBROOKLYN:TimeBinNight                         -0.638916   0.123874  -5.158
## BoroughMANHATTAN:TimeBinNight                         0.210729   0.172549   1.221
## BoroughQUEENS:TimeBinNight                           -0.329246   0.171626  -1.918
## BoroughSTATEN ISLAND:TimeBinNight                    -0.464996   0.303277  -1.533
## VictimAge18-24:TimeBinEvening                         0.055321   0.098504   0.562
## VictimAge25-44:TimeBinEvening                        -0.054011   0.096220  -0.561
## VictimAge65+:TimeBinEvening                          -1.431509   0.301603  -4.746
## VictimAgeNA:TimeBinEvening                           -0.420439   0.135214  -3.109
## VictimAge18-24:TimeBinMorning                         0.400581   0.201935   1.984
## VictimAge25-44:TimeBinMorning                         0.841688   0.192679   4.368
## VictimAge65+:TimeBinMorning                           0.157523   0.454076   0.347
## VictimAgeNA:TimeBinMorning                            0.732888   0.238493   3.073
## VictimAge18-24:TimeBinNight                           0.462698   0.105887   4.370
## VictimAge25-44:TimeBinNight                           0.370810   0.103658   3.577
## VictimAge65+:TimeBinNight                            -1.650125   0.388407  -4.248
## VictimAgeNA:TimeBinNight                             -0.276253   0.147286  -1.876
## BoroughBROOKLYN:VictimAge18-24:TimeBinEvening         0.188666   0.127643   1.478
## BoroughMANHATTAN:VictimAge18-24:TimeBinEvening       -0.100301   0.187741  -0.534
## BoroughQUEENS:VictimAge18-24:TimeBinEvening           0.106340   0.180470   0.589
## BoroughSTATEN ISLAND:VictimAge18-24:TimeBinEvening    0.095737   0.322455   0.297
## BoroughBROOKLYN:VictimAge25-44:TimeBinEvening         0.344820   0.124950   2.760
## BoroughMANHATTAN:VictimAge25-44:TimeBinEvening        0.005438   0.181168   0.030
## BoroughQUEENS:VictimAge25-44:TimeBinEvening           0.297971   0.175716   1.696
## BoroughSTATEN ISLAND:VictimAge25-44:TimeBinEvening    0.631634   0.322645   1.958
## BoroughBROOKLYN:VictimAge65+:TimeBinEvening           0.737591   0.392876   1.877
## BoroughMANHATTAN:VictimAge65+:TimeBinEvening          0.265366   0.547966   0.484
## BoroughQUEENS:VictimAge65+:TimeBinEvening             1.493286   0.551980   2.705
## BoroughSTATEN ISLAND:VictimAge65+:TimeBinEvening      2.399093   1.186284   2.022
## BoroughBROOKLYN:VictimAgeNA:TimeBinEvening            0.582110   0.176843   3.292
## BoroughMANHATTAN:VictimAgeNA:TimeBinEvening          -0.350735   0.240110  -1.461
## BoroughQUEENS:VictimAgeNA:TimeBinEvening              0.535706   0.243494   2.200
## BoroughSTATEN ISLAND:VictimAgeNA:TimeBinEvening       0.755501   0.420131   1.798
## BoroughBROOKLYN:VictimAge18-24:TimeBinMorning        -0.010227   0.269480  -0.038
## BoroughMANHATTAN:VictimAge18-24:TimeBinMorning       -0.220067   0.370717  -0.594
## BoroughQUEENS:VictimAge18-24:TimeBinMorning           0.689539   0.408291   1.689
## BoroughSTATEN ISLAND:VictimAge18-24:TimeBinMorning    1.044235   0.807011   1.294
```

```
## BoroughBROOKLYN:VictimAge25-44:TimeBinMorning       0.424956   0.255100   1.666
## BoroughMANHATTAN:VictimAge25-44:TimeBinMorning     -0.058756   0.347725  -0.169
## BoroughQUEENS:VictimAge25-44:TimeBinMorning         0.662389   0.396070   1.672
## BoroughSTATEN ISLAND:VictimAge25-44:TimeBinMorning  0.990893   0.799766   1.239
## BoroughBROOKLYN:VictimAge65+:TimeBinMorning         0.492965   0.597683   0.825
## BoroughMANHATTAN:VictimAge65+:TimeBinMorning        0.981269   0.703289   1.395
## BoroughQUEENS:VictimAge65+:TimeBinMorning           1.784809   0.780131   2.288
## BoroughSTATEN ISLAND:VictimAge65+:TimeBinMorning    3.466818   1.442050   2.404
## BoroughBROOKLYN:VictimAgeNA:TimeBinMorning          0.814347   0.308728   2.638
## BoroughMANHATTAN:VictimAgeNA:TimeBinMorning        -0.056896   0.409592  -0.139
## BoroughQUEENS:VictimAgeNA:TimeBinMorning            1.151381   0.452499   2.544
## BoroughSTATEN ISLAND:VictimAgeNA:TimeBinMorning     0.812012   0.908870   0.893
## BoroughBROOKLYN:VictimAge18-24:TimeBinNight         0.330094   0.141284   2.336
## BoroughMANHATTAN:VictimAge18-24:TimeBinNight        0.027657   0.198827   0.139
## BoroughQUEENS:VictimAge18-24:TimeBinNight           0.452812   0.192982   2.346
## BoroughSTATEN ISLAND:VictimAge18-24:TimeBinNight    0.407517   0.347387   1.173
## BoroughBROOKLYN:VictimAge25-44:TimeBinNight         0.566874   0.138504   4.093
## BoroughMANHATTAN:VictimAge25-44:TimeBinNight        0.079419   0.192775   0.412
## BoroughQUEENS:VictimAge25-44:TimeBinNight           0.546223   0.189068   2.889
## BoroughSTATEN ISLAND:VictimAge25-44:TimeBinNight    0.878323   0.348640   2.519
## BoroughBROOKLYN:VictimAge65+:TimeBinNight           0.908899   0.505421   1.798
## BoroughMANHATTAN:VictimAge65+:TimeBinNight         -0.241500   0.772254  -0.313
## BoroughQUEENS:VictimAge65+:TimeBinNight             1.310075   0.680796   1.924
## BoroughSTATEN ISLAND:VictimAge65+:TimeBinNight      2.426654   1.316939   1.843
## BoroughBROOKLYN:VictimAgeNA:TimeBinNight            0.788448   0.195147   4.040
## BoroughMANHATTAN:VictimAgeNA:TimeBinNight          -0.500823   0.263645  -1.900
## BoroughQUEENS:VictimAgeNA:TimeBinNight              0.492149   0.264196   1.863
## BoroughSTATEN ISLAND:VictimAgeNA:TimeBinNight       0.295096   0.484085   0.610
##                                                    Pr(>|z|)
## (Intercept)                                         < 2e-16 ***
## BoroughBROOKLYN                                    4.46e-09 ***
## BoroughMANHATTAN                                   3.17e-13 ***
## BoroughQUEENS                                      1.84e-09 ***
## BoroughSTATEN ISLAND                                < 2e-16 ***
## VictimAge18-24                                      < 2e-16 ***
## VictimAge25-44                                      < 2e-16 ***
## VictimAge65+                                        < 2e-16 ***
## VictimAgeNA                                        0.003641 **
## TimeBinEvening                                      < 2e-16 ***
## TimeBinMorning                                      < 2e-16 ***
## TimeBinNight                                       3.48e-05 ***
## BoroughBROOKLYN:VictimAge18-24                     0.519095
## BoroughMANHATTAN:VictimAge18-24                    0.889476
## BoroughQUEENS:VictimAge18-24                       0.519087
## BoroughSTATEN ISLAND:VictimAge18-24                0.498120
## BoroughBROOKLYN:VictimAge25-44                     0.124608
## BoroughMANHATTAN:VictimAge25-44                    0.582398
## BoroughQUEENS:VictimAge25-44                       0.469687
## BoroughSTATEN ISLAND:VictimAge25-44                0.036134 *
## BoroughBROOKLYN:VictimAge65+                       0.166918
## BoroughMANHATTAN:VictimAge65+                      0.852000
## BoroughQUEENS:VictimAge65+                         0.128425
## BoroughSTATEN ISLAND:VictimAge65+                  0.178320
## BoroughBROOKLYN:VictimAgeNA                        0.058503 .
```

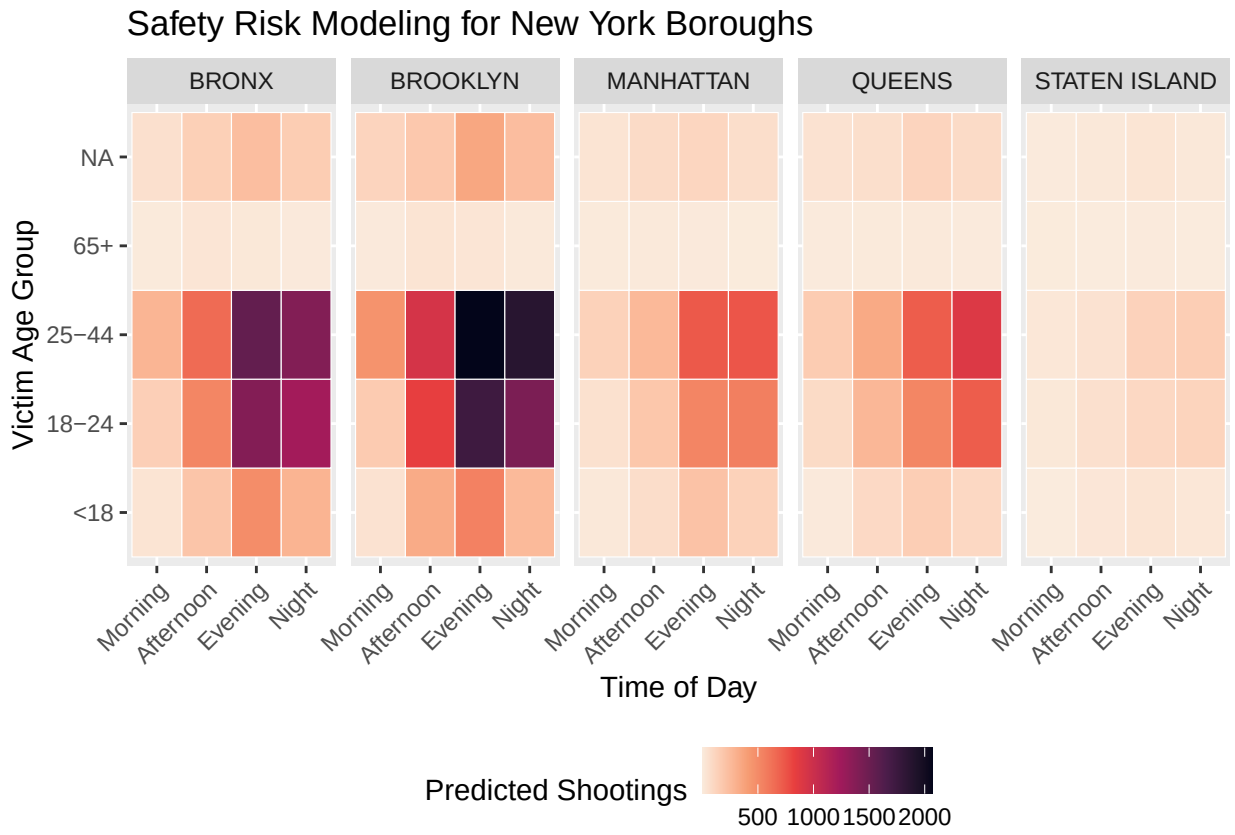
```
## BoroughMANHATTAN:VictimAgeNA                          0.021446 *
## BoroughQUEENS:VictimAgeNA                             0.838664
## BoroughSTATEN ISLAND:VictimAgeNA                      0.705586
## BoroughBROOKLYN:TimeBinEvening                        0.000118 ***
## BoroughMANHATTAN:TimeBinEvening                       0.314266
## BoroughQUEENS:TimeBinEvening                          0.008254 **
## BoroughSTATEN ISLAND:TimeBinEvening                   0.069650 .
## BoroughBROOKLYN:TimeBinMorning                        0.196043
## BoroughMANHATTAN:TimeBinMorning                       0.496156
## BoroughQUEENS:TimeBinMorning                          0.143645
## BoroughSTATEN ISLAND:TimeBinMorning                   0.262005
## BoroughBROOKLYN:TimeBinNight                          2.50e-07 ***
## BoroughMANHATTAN:TimeBinNight                         0.221985
## BoroughQUEENS:TimeBinNight                            0.055062 .
## BoroughSTATEN ISLAND:TimeBinNight                     0.125218
## VictimAge18-24:TimeBinEvening                         0.574383
## VictimAge25-44:TimeBinEvening                         0.574570
## VictimAge65+:TimeBinEvening                           2.07e-06 ***
## VictimAgeNA:TimeBinEvening                            0.001874 **
## VictimAge18-24:TimeBinMorning                         0.047288 *
## VictimAge25-44:TimeBinMorning                         1.25e-05 ***
## VictimAge65+:TimeBinMorning                           0.728659
## VictimAgeNA:TimeBinMorning                            0.002119 **
## VictimAge18-24:TimeBinNight                           1.24e-05 ***
## VictimAge25-44:TimeBinNight                           0.000347 ***
## VictimAge65+:TimeBinNight                             2.15e-05 ***
## VictimAgeNA:TimeBinNight                              0.060707 .
## BoroughBROOKLYN:VictimAge18-24:TimeBinEvening         0.139389
## BoroughMANHATTAN:VictimAge18-24:TimeBinEvening        0.593167
## BoroughQUEENS:VictimAge18-24:TimeBinEvening           0.555701
## BoroughSTATEN ISLAND:VictimAge18-24:TimeBinEvening    0.766543
## BoroughBROOKLYN:VictimAge25-44:TimeBinEvening         0.005786 **
## BoroughMANHATTAN:VictimAge25-44:TimeBinEvening        0.976054
## BoroughQUEENS:VictimAge25-44:TimeBinEvening           0.089933 .
## BoroughSTATEN ISLAND:VictimAge25-44:TimeBinEvening    0.050268 .
## BoroughBROOKLYN:VictimAge65+:TimeBinEvening           0.060461 .
## BoroughMANHATTAN:VictimAge65+:TimeBinEvening          0.628192
## BoroughQUEENS:VictimAge65+:TimeBinEvening             0.006824 **
## BoroughSTATEN ISLAND:VictimAge65+:TimeBinEvening      0.043139 *
## BoroughBROOKLYN:VictimAgeNA:TimeBinEvening            0.000996 ***
## BoroughMANHATTAN:VictimAgeNA:TimeBinEvening           0.144091
## BoroughQUEENS:VictimAgeNA:TimeBinEvening              0.027802 *
## BoroughSTATEN ISLAND:VictimAgeNA:TimeBinEvening       0.072137 .
## BoroughBROOKLYN:VictimAge18-24:TimeBinMorning         0.969727
## BoroughMANHATTAN:VictimAge18-24:TimeBinMorning        0.552762
## BoroughQUEENS:VictimAge18-24:TimeBinMorning           0.091249 .
## BoroughSTATEN ISLAND:VictimAge18-24:TimeBinMorning    0.195681
## BoroughBROOKLYN:VictimAge25-44:TimeBinMorning         0.095744 .
## BoroughMANHATTAN:VictimAge25-44:TimeBinMorning        0.865818
## BoroughQUEENS:VictimAge25-44:TimeBinMorning           0.094444 .
## BoroughSTATEN ISLAND:VictimAge25-44:TimeBinMorning    0.215353
## BoroughBROOKLYN:VictimAge65+:TimeBinMorning           0.409489
## BoroughMANHATTAN:VictimAge65+:TimeBinMorning          0.162939
## BoroughQUEENS:VictimAge65+:TimeBinMorning             0.022147 *
```

```
## BoroughSTATEN ISLAND:VictimAge65+:TimeBinMorning   0.016213 *
## BoroughBROOKLYN:VictimAgeNA:TimeBinMorning         0.008346 **
## BoroughMANHATTAN:VictimAgeNA:TimeBinMorning        0.889522
## BoroughQUEENS:VictimAgeNA:TimeBinMorning           0.010944 *
## BoroughSTATEN ISLAND:VictimAgeNA:TimeBinMorning    0.371627
## BoroughBROOKLYN:VictimAge18-24:TimeBinNight        0.019472 *
## BoroughMANHATTAN:VictimAge18-24:TimeBinNight       0.889372
## BoroughQUEENS:VictimAge18-24:TimeBinNight          0.018956 *
## BoroughSTATEN ISLAND:VictimAge18-24:TimeBinNight   0.240759
## BoroughBROOKLYN:VictimAge25-44:TimeBinNight        4.26e-05 ***
## BoroughMANHATTAN:VictimAge25-44:TimeBinNight       0.680355
## BoroughQUEENS:VictimAge25-44:TimeBinNight          0.003864 **
## BoroughSTATEN ISLAND:VictimAge25-44:TimeBinNight   0.011759 *
## BoroughBROOKLYN:VictimAge65+:TimeBinNight          0.072129 .
## BoroughMANHATTAN:VictimAge65+:TimeBinNight         0.754493
## BoroughQUEENS:VictimAge65+:TimeBinNight            0.054313 .
## BoroughSTATEN ISLAND:VictimAge65+:TimeBinNight     0.065381 .
## BoroughBROOKLYN:VictimAgeNA:TimeBinNight           5.34e-05 ***
## BoroughMANHATTAN:VictimAgeNA:TimeBinNight          0.057484 .
## BoroughQUEENS:VictimAgeNA:TimeBinNight             0.062487 .
## BoroughSTATEN ISLAND:VictimAgeNA:TimeBinNight      0.542129
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 4.9224e+04  on 99  degrees of freedom
## Residual deviance: 9.3348e-13  on  0  degrees of freedom
## AIC: 832.43
##
## Number of Fisher Scoring iterations: 3
```

I can't follow that much data visually to get any interpretations out of it. Let's get this info into a heatmap to understand it better.

```
heatmap_data <- safety_matrix %>%
  mutate(Predicted = predict(mod, type="response")) %>%
  mutate(TimeBin = factor(TimeBin, levels = c("Morning", "Afternoon", "Evening", "Night")))

ggplot(heatmap_data, aes(x = TimeBin, y = VictimAge, fill = Predicted)) +
  geom_tile(color = "white", linewidth = 0.2) +
  facet_wrap(~Borough, scales = "free_x", ncol = 5) +
  scale_fill_viridis_c(option = "rocket", direction = -1, name="Predicted Shootings") +
  labs(
    title = "Safety Risk Modeling for New York Boroughs",
    x = "Time of Day",
    y = "Victim Age Group"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "bottom")
```



Analysis of Modeling Heatmap

Now we're talking! It looks like my parents were right! Be careful of venturing out in Brooklyn or the Bronx late at night as the predicted shootings where adults are the victims rises greatly. Manhattan and Queens aren't incredibly safe, either.

It is interesting that older adults at 65+ are not the victims of shootings terribly often. Makes sense that middle-aged adults, who might own and have easy access to guns, would be the ones in the most danger. It is terrible to think of the losses in the under 18 category. Too often we hear about gun owners not securing their weapons, not having proper training, or kids using real weapons as toys. We can't say precisely what happened in these cases, but the data and numbers do seem to line up with real world scenarios.

Possible Bias

There is always a degree of bias in this data, intentional or not. In this case, I do not really know how NYPD pulls these numbers. I imagine there are plenty of scenarios where there are shootings that NYPD is not made aware of. Even in cleaning up this data, I adjusted a few numbers that appeared to have been entered incorrectly due to human error.

When pulling this data, it mentioned the information stopped at the end of 2024 because they expanded some of the data fields. I remember it mentioned the new report allowed to account for shooting events with multiple victims. I didn't dive into this data too granularly to see, but how would a mass shooting event be identified in this report? Would there be multiple lines for multiple victims shot? Is it all just on one line, where we would lose the age, sex, and race data for more than one victim? I do think this data gives a pretty

good, general idea of shootings in New York City. I am sure it is not a perfect representation, though, so it is very possible some missing information and context is swaying these numbers slightly away from reality.